

21/05/25 Ex:-5 :- Study of Activation Functions And their Role.

Aim: To study the different activation functions used in neural networks and analyze their role in introducing non-linearity, enabling the network to learn complex patterns.

* Objectives :-

1. To Understand the Mathematical behaviour of commonly used activation functions.
2. To implement Sigmoid, Tanh, Relu, and Softmax functions.
3. To Compare the effect of different activation functions on model Performance.
4. To observe the importance of non-linearity in neural network.

* Pseudo Code:-

START

Import necessary Libraries

Define activation functions.

$$\text{Sigmoid}(x) = \frac{1}{1 + \exp(-x)}$$

$$\text{Tanh}(x) = \frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)}$$

$$\text{Relu}(x) = \max(0, x)$$

$$\text{softmax}(x) = \frac{\exp(x)}{\sum \exp(x)}$$

load dataset (MNIST)

For each activation function in [sigmoid, Tanh, Relu],

Build a neural network with that activation

Train the network on training data

Evaluate on test data

store accuracy

Compare accuracy results

END

* Observations :-

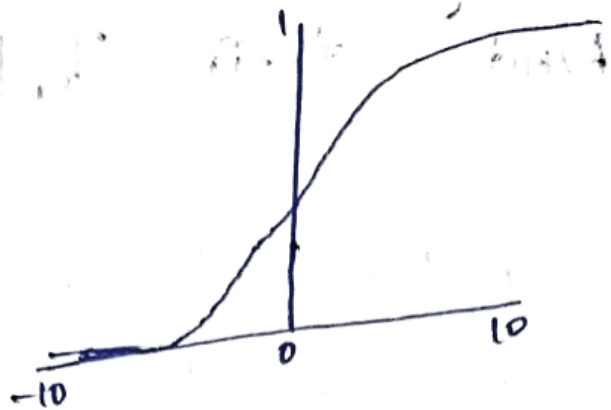
- 1, Sigmoid :- smooth, but suffers from Vanishing gradient for large positive/negative inputs
- 2, Tanh :- Better than sigmoid, outputs range $(-1, 1)$ but still faces Vanishing gradient
- 3, Relu :- Most effective in deep networks, avoids Vanishing gradient, fast convergence
- 4, SoftMax :- Used in output layer for Multi-classification Problems.

* Conclusion :-

Activation functions introduce non-linearity into neural networks, enabling them to approximate complex mapping between inputs and outputs.

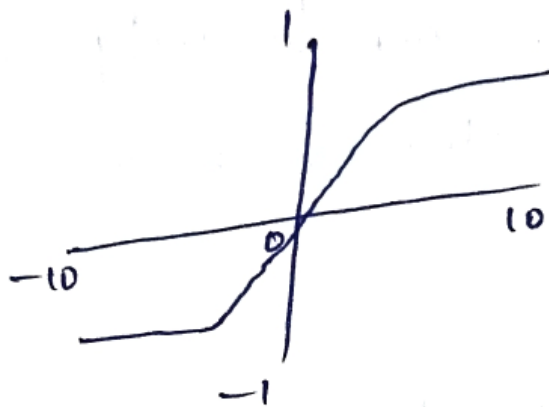
Sigmoid:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



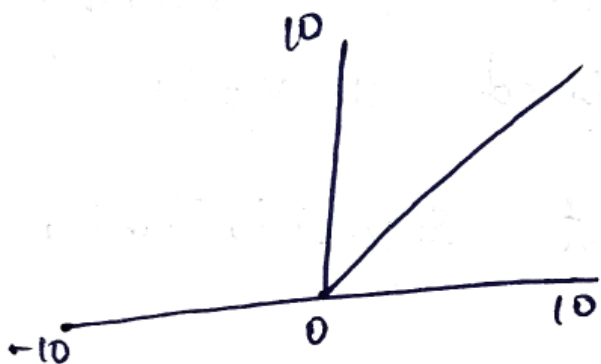
Tanh:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Relu

$$g(z) = \max(0, z)$$



Result:

Successfully Implemented Activation functions
and its different roles. [Sigmoid, Tanh, ReLU]